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For general purpose parallel Python work, what should we prefer?

Multi-threading

0%

Multi-processing

0%

Multi-sockets

0%



59

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Preparing Results

For general purpose parallel Python work, what should we prefer?

Multi-threading



70%

Multi-processing



30%

Multi-sockets



0%

RESULTS SLIDE

When can we use multi-threading with good speed-ups?

When the work is very small

0%

When the stuff we call releases the GIL

0%

Always

0%



73

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Preparing Results



When can we use multi-threading with good speed-ups?

When the work is very small



2.74%

When the stuff we call releases the GIL



95.89%

Always



1.37%

RESULTS SLIDE



Why is multi-processing not limited by the GIL?

Each process has its own GIL

0%

It turns it off

0%

Processes manually release the GIL

0%



77

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Preparing Results



Why is multi-processing not limited by the GIL?

Each process has its own GIL



It turns it off



Processes manually release the GIL



RESULTS SLIDE



For parallel workloads we expect the wall time to

Go down

0%

Stay approximately constant

0%

Go up

0%



74

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Preparing Results



For parallel workloads we expect the wall time to

Go down



96%

Stay approximately constant



4%

Go up



0%

RESULTS SLIDE



For parallel workloads we expect the CPU time to

Go down

0%

Stay approximately constant

0%

Go up

0%



73

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Preparing Results



For parallel workloads we expect the CPU time to

Go down



2.74%

Stay approximately constant



89.04%

Go up



8.22%

RESULTS SLIDE

What is (roughly) the speed-up between 1 and 2 threads?

0.6		0%
1.6		0%
2.0		0%

```
$ # Single thread
$ time python sums.py
```

```
real    0m2.916s
```

```
user    0m2.800s
```

```
sys     0m0.069s
```

```
$ # 2 threads
```

```
$ time python sums.py
```

```
real    0m1.855s
```

```
user    0m2.890s
```

```
sys     0m0.048s
```



72

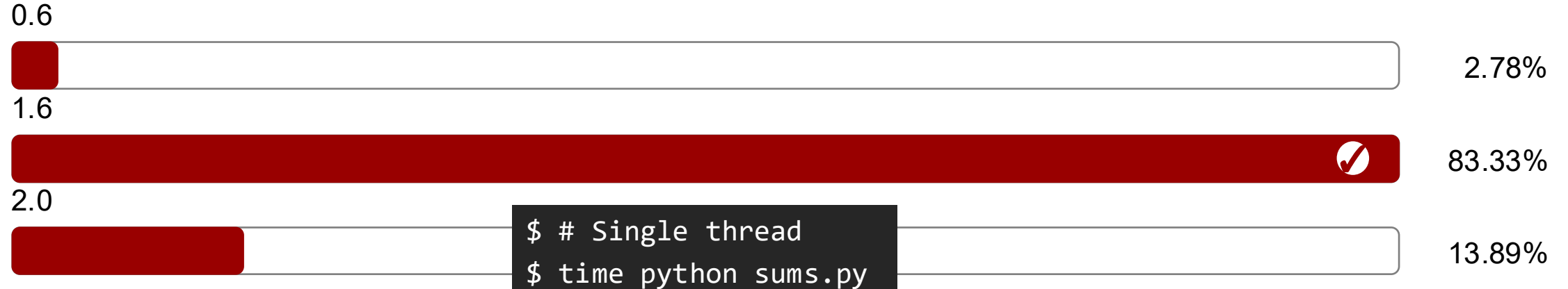
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Preparing Results



What is (roughly) the speed-up between 1 and 2 threads?



```
$ # Single thread  
$ time python sums.py
```

```
real    0m2.916s  
user    0m2.800s  
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```

RES

SLIDE

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How do we solve overhead with small work tasks?

Chunk tasks together

0%

Switch to multi-threading to reduce overhead

0%

Add more tasks

0%



66

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Preparing Results



How do we solve overhead with small work tasks?

Chunk tasks together



97.01%

Switch to multi-threading to reduce overhead



2.99%

Add more tasks



0%

RESULTS SLIDE



Why might static scheduling cause issues with load balancing?

Processes may refuse tasks so they have to be re-done afterwards

0%

Too many tasks may be scheduled to one process

0%

Some tasks may take much longer than others

0%



73

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Preparing Results

Why might static scheduling cause issues with load balancing?

Processes may refuse tasks so they have to be re-done afterwards



1.37%

Too many tasks may be scheduled to one process



2.74%

Some tasks may take much longer than others



95.89%

RESULTS SLIDE



**Assume we have a program with parallel fraction $F = 0.9$.
What is (roughly) the expected speed-up with 4 cores?**

2

0%

3

0%

4

0%

$$S(p) = \frac{1}{(1 - F) + \frac{F}{p}}$$

$$S(\infty) = \frac{1}{1 - F} = \frac{1}{B}$$



Assume we have a program with parallel fraction $F = 0.9$.
What is (roughly) the expected speed-up with 4 cores?

2



2.86%

3



74.29%

4



22.86%

$$S(p) = \frac{1}{(1 - F) + \frac{F}{p}}$$

$$S(\infty) = \frac{1}{1 - F} = \frac{1}{B}$$

RESULTS SLIDE



**Assume we have a program with parallel fraction $F = 0.9$.
What is the theoretical maximum speed-up?**

10

0%

100

0%

1000

0%

$$S(p) = \frac{1}{(1 - F) + \frac{F}{p}}$$

$$S(\infty) = \frac{1}{1 - F} = \frac{1}{F}$$



Assume we have a program with parallel fraction $F = 0.9$.
What is the theoretical maximum speed-up?

10

☒

100%

100

☐

0%

1000

☐

0%

$$S(p) = \frac{1}{(1 - F) + \frac{F}{p}}$$

$$S(\infty) = \frac{1}{1 - F} = \frac{1}{B}$$

RESULTS SLIDE

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